Model of the dynamics of *Diaphorina citri* in the presence of a predator

$$\begin{aligned}
D'(t) &= \phi D(B - \varphi D) - (\delta + \nu P)D, \\
P'(t) &= \lambda PD - \gamma P \\
B'(t) &= \alpha BS - \mu B - \varphi_D BD, \\
S'(t) &= \rho S\left(1 - \frac{S}{K}\right) - \beta BS,
\end{aligned}$$

where initial conditions are $D(0) = D, P(0) = P_0, B(0) = B_0$ y $S_0(0) = S_0$ and the variables are

- $D(t)$ is the average number of individuals of *Diaphorina citri* at time t,
- $P(t)$ is the average number of individuals of Tamarixia radiata, predator of *Diaphorina citri,* at time t,
- $B(t)$ is the average number of shoots on the host crop plant at time t,
- $S(t)$ is intrinsic plant stimulus responsible for the sprouting process at time t.

Par	**Description**
ϕ	XXXXXXXXXX adult of *Diaphorina citri* per shoot
$1/\varphi$	Average number of *Diaphorina citri* per shoot
ν	*Diaphorina citri* mortality rate due to predation
λ	growth rate of Tamarixia radiata by consumption of *Diaphorina citri*
δ	*Diaphorina citri* mortality rate
θ	Tamarixia radiata mortality rate
ρ	Intrinsic growth rate of plant shoots
α	growth rate of the shoots due to the stimulus
β	rate of decrease in stimulus due to sprouting
μ	transition rate from shoot to branch
φ_D	Rate of affectation to the shoot by occupation
K	Shoot carrying capacity per plant